

Spectral Feature Augmentation for Graph Contrastive Learning and Beyond

Yifei Zhang¹, Hao Zhu^{2,3}, Zixing Song¹, Piotr Koniusz^{3,2,*}, Irwin King¹

¹The Chinese University of Hong Kong, ²Australian National University, ³Data61/CSIRO
 {yfzhang, zxsong, king}@cse.cuhk.edu.hk;
 allenhaozhu@gmail.com; piotr.koniusz@data61.csiro.au

Abstract

Although augmentations (*e.g.*, perturbation of graph edges, image crops) boost the efficiency of Contrastive Learning (CL), feature level augmentation is another plausible, complementary yet not well researched strategy. Thus, we present a novel spectral feature argumentation for contrastive learning on graphs (and images). To this end, for each data view, we estimate a low-rank approximation per feature map and subtract that approximation from the map to obtain its complement. This is achieved by the proposed herein incomplete power iteration, a non-standard power iteration regime which enjoys two valuable byproducts (under mere one or two iterations): (i) it partially balances spectrum of the feature map, and (ii) it injects the noise into rebalanced singular values of the feature map (spectral augmentation). For two views, we align these rebalanced feature maps as such an improved alignment step can focus more on less dominant singular values of matrices of both views, whereas the spectral augmentation does not affect the spectral angle alignment (singular vectors are not perturbed). We derive the analytical form for: (i) the incomplete power iteration to capture its spectrum-balancing effect, and (ii) the variance of singular values augmented implicitly by the noise. We also show that the spectral augmentation improves the generalization bound. Experiments on graph/image datasets show that our spectral feature augmentation outperforms baselines, and is complementary with other augmentation strategies and compatible with various contrastive losses.

1 Introduction

Semi-supervised and supervised Graph Neural Networks (GNNs) (???) require full access to class labels. However, unsupervised GNNs (???) and recent Self-Supervised Learning (SSL) models do not require labels (??) to train embeddings. Among SSL methods, Contrastive Learning (CL) achieves comparable performance with its supervised counterparts on many tasks (??). CL has also been applied recently to the graph domain. A typical Graph Contrastive Learning (GCL) method forms multiple graph views via stochastic augmentation of the input to learn representations by contrasting so-called positive samples with negative samples (????). As an indispensable part of GCL, the significance of graph

*Corresponding author. PK was primarily concerned with the theoretical analysis (*e.g.*, Prop. 2 & 3). This paper has been published with the Thirty-Seventh AAAI Conference on Artificial Intelligence (AAAI 2023).

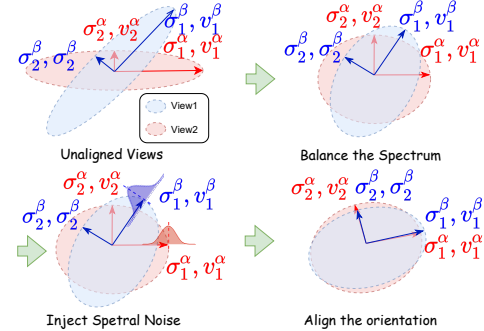
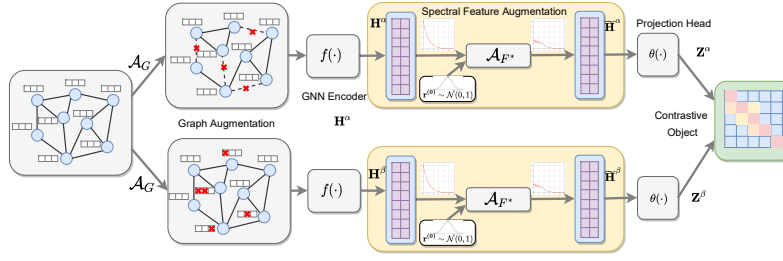


Figure 1: Our GCL with spectral feature augmentation by the incomplete power iteration implicitly performs three steps. Let blue and red ellipses represent spectra of feature maps H^α and H^β of two views. Singular values of unaligned views (top left) are firstly rebalanced (top right) and the noise is injected into singular values (bottom left). Given rebalanced singular values, corresponding singular vectors are aligned with equalized emphasis (leading unbalanced singular values would emphasize the alignment of leading singular vectors, sacrificing the quality of alignment of other singular vectors).

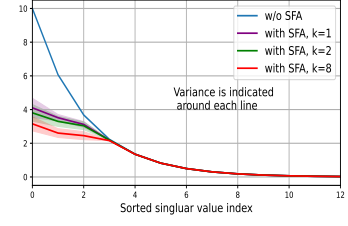
augmentation has been well studied (???). Popular random data augmentations are just one strategy to construct views, and their noise may affect adversely downstream tasks (??). Thus, some works (???) learn graph augmentations but they require supervision.

The above issue motivates us to propose a simple/efficient data augmentation model which is complementary with existing augmentation strategies. We target Feature Augmentation (FA) as scarcely any FA works exist in the context of CL and GCL. In the image domain, a simple FA (??) showed that perturbing feature representations of an image results in a representation of another image where both images share some semantics (?). However, perturbing features randomly ignores covariance of feature representations, and ignores semantics correlations. Hence, we opt for injecting random noise into the singular values of feature maps as such a spectral feature augmentation does not alter the orthogonal bases of feature maps by much, thus helping preserve semantics correlations.

Moreover, as typical GCL aligns two data views (?), unbalanced singular values of two data views may affect the quality of alignment. As several leading singular values (acting as



(a) Our pipeline.



(b) Simulation: spectrum obtained by Alg. 1.

Figure 2: Our GCL model. Two graph views are generated by data augmentation and passed into graph neural network encoders with shared parameters to learn node representations. The proposed spectral feature augmentation rebalances (partially equalizes) the spectrum of each feature map, and implicitly injects the noise into rebalanced singular values. Such representations are fed into the projection head and the contrastive loss. Figure 1 explains the role of our spectral feature augmentation.

weights on the loss) dominate the alignment process, GCL favors aligning the leading singular vectors of two data views while sacrificing remaining orthogonal directions with small singular values. In other words, the unbalanced spectrum leads to a suboptimal orthonormal bases alignment, which results in a suboptimal GCL model.

To address rebalancing of unbalanced spectrum and augmenting leading singular values, we present a novel and efficient *Spectral Feature Augmentation* (SFA). To this end, we propose the so-called incomplete power iteration which, under just one or two iterations, partially balances singular values of feature maps and implicitly injects the noise into these singular values. We evaluate our method on various datasets for node level tasks (*i.e.*, node classification and node clustering). We also show that our method is compatible with other augmentation strategies and contrastive losses.

We summarize our contributions as follows:

- i. We propose a simple/efficient spectral feature augmentation for GCL which is independent of different contrastive losses, *i.e.*, we employ InfoNCE and Barlow Twin.
- ii. We introduce the so-called incomplete power iteration which, under just one or two iterations, partially balances spectra of two data views and injects the augmentation noise into their singular values. The rebalanced spectra help align orthonormal bases of both data views.
- iii. As the incomplete power iteration is stochastic in its nature, we derive its analytical form which provably demonstrates its spectrum rebalancing effect in expectation, and captures the variance of the spectral augmentation.
- iv. For completeness, we devise other spectral augmentation models, based on the so-called MaxExp and Power Norm. operators and Grassman feature maps, whose rebalancing and noise injection profiles differ with our method.

2 Related Work

Data Augmentation. Augmentations are usually performed in the input space. In computer vision, image transformations, *i.e.*, rotation, flipping, color jitters, translation, noise injection (?), cutout and random erasure (?) are popular. In neural language processing, token-level random augmentations, *e.g.*, synonym replacement, word swapping, word insertion, and

deletion (?) are used. In transportation, conditional augmentation of road junctions is used (?). In the graph domain, attribute masking, edge permutation, and node dropout are popular (?). Sun *et al.* (?) use adversarial graph perturbations. Zhu *et al.* (?) use adaptive graph augmentations based on the node/PageRank centrality (?) to mask edges with varying probability.

Feature Augmentation. Samples can be augmented in the feature space instead of the input space (?). Wang *et al.* (?) augment the hidden space features, resulting in auxiliary samples with the same class identity but different semantics. A so-called channel augmentation perturbs the channels of feature maps (?) while GCL approach, COSTA (?), augments features via random projections. Some few-shot learning approaches augment features (?) while others estimate the “analogy” transformations between samples of known classes to apply them on samples of novel classes (??) or mix foregrounds and backgrounds (?). However, “analogy” augmentations are not applicable to contrastive learning due to the lack of labels.

Graph Contrastive Learning. CL is popular in computer vision, NLP (???), and graph learning. In the vision domain, views are formed by augmentations at the pixel level, whereas in the graph domain, data augmentation may act on node attributes or the graph edges. GCL often explores node-node, node-graph, and graph-graph relations for contrastive loss which is similar to contrastive losses in computer vision. Inspired by SimCLR (?), GRACE (?) correlates graph views by pushing closer representations of the same node in different views and separating representations of different nodes, and Barlow Twin (?) avoids the so-called dimensional collapse (?).

In contrast, we study spectral feature augmentations to perturb/rebalance singular values of both views. We outperform feature augmentations such as COSTA (?).

3 Proposed Method

Inspired by recent advances in augmentation-based GCL, our approach learns node representations by rebalancing spectrum of two data views and performing the spectral feature augmentation via the incomplete power iteration. SFA is complementary to the existing data augmentation approaches.

Figure 2a illustrates our framework. The ?? section (supplementary material) explains our notations.

Graph Augmentation ($\mathcal{A}_G$). Augmented graph $(\tilde{\mathbf{A}}, \tilde{\mathbf{X}})$ is generated by $\mathcal{A}_G$ by directly adding random perturbations to the original graph $(\mathbf{A}, \mathbf{X})$. Different augmented graphs are constructed given one input $(\mathbf{A}, \mathbf{X})$, yielding correlated views, *i.e.*, $(\tilde{\mathbf{A}}^\alpha, \tilde{\mathbf{X}}^\alpha)$ and $(\tilde{\mathbf{A}}^\beta, \tilde{\mathbf{X}}^\beta)$. In the common GCL setting (?), the graph structure is augmented by permuting edges, whereas attributes by masking.

Graph Neural Network Encoders. Our framework admits various choices of the graph encoder. We opt for simplicity and adopt the commonly used graph convolution network (GCN) (?) as our base graph encoder. As shown in Fig. 2a, we use a shared graph encoder for each view, *i.e.*, $f: \mathbb{R}^{n \times d_x} \times \mathbb{R}^{n \times n} \mapsto \mathbb{R}^{n \times d_h}$. We consider two graphs generated from $\mathcal{A}_G$ as two congruent structural views and define the GCN encoder with 2 layers as:

$$f(\mathbf{X}, \mathbf{A}) = \text{GCN}_2(\text{GCN}_1(\mathbf{X}, \mathbf{A}), \mathbf{A}), \quad (1)$$

where $\text{GCN}_l(\mathbf{X}, \mathbf{A}) = \sigma(\hat{\mathbf{D}}^{-\frac{1}{2}} \hat{\mathbf{A}} \hat{\mathbf{D}}^{-\frac{1}{2}} \mathbf{X} \Theta)$.

Moreover, $\tilde{\mathbf{A}} = \hat{\mathbf{D}}^{-1/2} \hat{\mathbf{A}} \hat{\mathbf{D}}^{-1/2} \in \mathbb{R}^{n \times n}$ is the degree-normalized adjacency matrix, $\hat{\mathbf{D}} \in \mathbb{R}^{n \times n}$ is the degree matrix of $\hat{\mathbf{A}} = \mathbf{A} + \mathbf{I}_N$ where $\mathbf{I}_N$ is the identity matrix, $\mathbf{X} \in \mathbb{R}^{n \times d_x}$ contains the initial node features, $\Theta \in \mathbb{R}^{d_x \times d_h}$ contains network parameters, and $\sigma(\cdot)$ is a parametric ReLU (PReLU). The encoder outputs feature maps $\mathbf{H}^\alpha$ and $\mathbf{H}^\beta$ for two views.

Spectral Feature Augmentation (SFA). $\mathbf{H}^\alpha$ and $\mathbf{H}^\beta$ are fed to the feature augmenting function $\mathcal{A}_{F^*}$ where random noises are added to the spectrum via the incomplete power iteration. We explain the proposed SFA in the [Spectral Feature Augmentation for GCL](#) section and detail its properties in Propositions 1, 2 and 3. SFA results in the spectrally-augmented feature maps, *i.e.*, $\tilde{\mathbf{H}}^\alpha$ and $\tilde{\mathbf{H}}^\beta$. SFA is followed by a shared projection head $\theta: \mathbb{R}^{n \times d_h} \mapsto \mathbb{R}^{n \times d_z}$ which is an MLP with two hidden layers and PReLU nonlinearity. It maps $\tilde{\mathbf{H}}^\alpha$ and $\tilde{\mathbf{H}}^\beta$ into two node representations $\mathbf{Z}^\alpha, \mathbf{Z}^\beta \in \mathbb{R}^{n \times d_z}$ (two congruent views of one graph) on which the contrastive loss is applied. As described in (?), it is beneficial to define the contrastive loss on $\mathbf{Z}$ rather than $\mathbf{H}$.

Contrastive Training. To train the encoders end-to-end and learn rich node representations that are agnostic to downstream tasks, we utilize the InfoNCE loss (?):

$$\mathcal{L}_{\text{contrastive}}(\tau) = \underbrace{\mathbb{E}_{\mathbf{z}, \mathbf{z}^+} [-\mathbf{z}^\top \mathbf{z}^+ / \tau]}_{\text{alignment}} + \underbrace{\mathbb{E}_{\mathbf{z}, \mathbf{z}^+} \left[\log \left(e^{\mathbf{z}^\top \mathbf{z}^+ / \tau} + \sum_{\mathbf{z}^- \in \mathbf{Z}^{\alpha\beta} \setminus \{\mathbf{z}, \mathbf{z}^+\}} e^{\mathbf{z}^\top \mathbf{z}^- / \tau} \right) \right]}_{\text{uniformity}}, \quad (2)$$

where $\mathbf{z}$ is the representation of the anchor node in one view (*i.e.*, $\mathbf{z} \in \mathbf{Z}^\alpha$) and $\mathbf{z}^+$ denotes the representation of the anchor node in another view (*i.e.*, $\mathbf{z}^+ \in \mathbf{Z}^\beta$), whereas $\{\mathbf{z}^-\}$ are from the set of node representations other than $\mathbf{z}$ and $\mathbf{z}^+$ (*i.e.*, $\mathbf{Z}^{\alpha\beta} \equiv \mathbf{Z}^\alpha \cup \mathbf{Z}^\beta$ and $\mathbf{z}^- \in \mathbf{Z}^{\alpha\beta} \setminus \{\mathbf{z}, \mathbf{z}^+\}$). The first part of

Algorithm 1: Spectral Feature Augmentation ($\mathcal{A}_{F^*}$)

Input: feature map $\mathbf{H}$; the number of iterations k ;

$\mathbf{r}^{(0)} \sim \mathcal{N}(0, \mathbf{I})$

for $i = 1$ **to** k **do**

$\mathbf{r}^{(i)} = \mathbf{H}^\top \mathbf{H} \mathbf{r}^{(i-1)}$

end for

$\tilde{\mathbf{H}} = \mathbf{H} - \frac{\mathbf{H} \mathbf{r}^{(k)} \mathbf{r}^{(k)\top}}{\|\mathbf{r}^{(k)}\|_2^2}$

Return $\tilde{\mathbf{H}}$

Eq. (2) maximizes the alignment of two views (representations of the same node become similar). The second part of Eq. (2) minimizes the pairwise similarity via LogSumExp. Pushing node representations away from each other makes them uniformly distributed (?).

3.1 Spectral Feature Augmentation for GCL

Our spectral feature augmentation is inspired by the rank-1 update (?). Let $\mathbf{H} = f(\mathbf{X}, \mathbf{A})$ be the graph feature map with the singular decomposition $\mathbf{H} = \mathbf{U} \Sigma \mathbf{V}^\top$ where $\mathbf{H} \in \mathbb{R}^{n \times d_h}$, $\mathbf{U}$ and $\mathbf{V}$ are unitary matrices, and $\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_{d_h})$ is the diagonal matrix with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{d_h}$. Starting from a random point $\mathbf{r}^{(0)} \sim \mathcal{N}(0, \mathbf{I})$ and function $\mathbf{r}^{(k)} = \mathbf{H}^\top \mathbf{H} \mathbf{r}^{(k-1)}$, we generate a set of augmented feature maps¹ $\tilde{\mathbf{H}}$ by:

$$\tilde{\mathbf{H}}(\mathbf{H}; \mathbf{r}^{(0)}) = \mathbf{H} - \mathbf{H}_{\text{LowRank}} = \mathbf{H} - \frac{\mathbf{H} \mathbf{r}^{(k)} \mathbf{r}^{(k)\top}}{\|\mathbf{r}^{(k)}\|_2^2}. \quad (3)$$

We often write $\tilde{\mathbf{H}}$ rather than $\tilde{\mathbf{H}}(\mathbf{H}; \mathbf{r}^{(0)})$, and we often think of $\tilde{\mathbf{H}}$ as a matrix. We summarize the proposed SFA in Alg. 1.

Proposition 1. *Let $\tilde{\mathbf{H}}$ be the augmented feature matrix obtained via Alg. 1 for the k -th iteration starting from a random vector $\mathbf{r}^{(0)}$ drawn from $\mathcal{N}(0, \mathbf{I})$. Then $\mathbb{E}_{\mathbf{r}^{(0)} \sim \mathcal{N}(0, \mathbf{I})}(\tilde{\mathbf{H}}(\mathbf{H}; \mathbf{r}^{(0)})) = \mathbf{U} \tilde{\Sigma} \mathbf{V}^\top$ has rebalanced spectrum² $\tilde{\Sigma} = \text{diag}[(1 - \lambda_1(k))\sigma_1, (1 - \lambda_2(k))\sigma_2, \dots, (1 - \lambda_{d_h}(k))\sigma_{d_h}]$ where $\lambda_i(k) = \mathbb{E}_{\mathbf{y} \sim \mathcal{N}(0, \mathbf{I})} \left(\frac{(y_i \sigma_i^{2k})^2}{\sum_{l=1}^{d_h} (y_l \sigma_l^{2k})^2} \right)$ and $\mathbf{y} = \mathbf{V}^\top \mathbf{r}^{(0)}$, because $0 \leq 1 - \lambda_1(k) \leq 1 - \lambda_2(k) \leq \dots \leq 1 - \lambda_{d_h}(k) \leq 1$ for $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{d_h}$ (sorted singular values from the SVD), and so $(1 - \lambda_i)$ gets smaller or larger as σ_i gets larger or smaller, respectively.*

Push-forward Function (Partial balancing of spectrum). Prop. 1 shows that in expectation, our incomplete power iteration rebalances spectrum according to the push-forward function $\phi(\sigma_i; k) = \sigma_i(1 - \lambda_i(k))$ where $\lambda_i(k)$ is an expected value of $\lambda'_i(\mathbf{y}, k) = \frac{(y_i \sigma_i^{2k})^2}{\sum_{l=1}^{d_h} (y_l \sigma_l^{2k})^2}$ w.r.t. random variable $\mathbf{y} \sim \mathcal{N}(0, \mathbf{I})$ (see Eq. (??)).

¹We apply Eq. (3) on both views $\mathbf{H}^\alpha$ and $\mathbf{H}^\beta$ separately to obtain spectrally rebalanced/augmented $\tilde{\mathbf{H}}^\alpha$ and $\tilde{\mathbf{H}}^\beta$.

²“Rebalanced” means the output spectrum is flatter than the input.

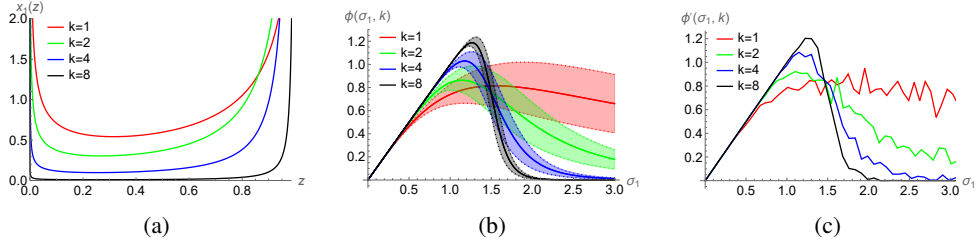


Figure 3: Toy illustration of Prop. 2 and 3. Let $\sigma_2, \dots, \sigma_5$ be 1.5, 0.9, 0.2, 0.01. We investigate the impact of iterations $k \in \{1, 2, 4, 8\}$. Fig. 3a shows distribution $x(z)$ given $\sigma_1 = 2$. Fig. 3b shows the expected value $\phi(\sigma_1, k) = \sigma_1(1 - \lambda_1)$ where $\lambda_1 = \mathbb{E}_{z \sim x_1}(z)$ for $0 \leq \sigma_1 \leq 3$. The deviation is indicated by $\phi_{\pm \omega_1}(\sigma_1, k) = \sigma_1(1 - \lambda_1 \pm \omega_1)$. Finally, Fig. 3c is obtained via Alg. 1 (the incomplete power iteration). To this end, we generated randomly a feature matrix $\mathbf{H}$ and substituted its singular values by $\sigma_1, \dots, \sigma_5$. Notice that for $k = 1$, $1 \leq \sigma_1 \leq 3$, push-forward $\phi(\sigma_1, 1)$ and $\phi'(\sigma_1, 1)$ in Fig. 3b and 3c are around 0.8 (the balancing of spectrum) and the high deviation indicates the singular value undergoes the spectral augmentation. For $k \geq 2$, both balancing and spectral augmentation effects decline. Note theoretical ϕ in Prop. 2 and real ϕ' from Alg. 1 match.

Alg. 1 returns an instance governed by the currently drawn $\mathbf{y}$. The push-forward function in a feed-forward step of network realizes $\phi'(\sigma_i; \mathbf{y}, k) = \sigma_i(1 - \lambda'_i(\mathbf{y}, k))$. Thus, below we study the analytical expression for $\phi(\sigma_i; k)$ and its variance to understand how drawing $\mathbf{y} \sim \mathcal{N}(0, \mathbf{I})$ translates into the variance posed by the implicit spectral augmentation of the singular values.

Proposition 2. Analytical Expectation. Let $\beta_i = (\sigma_i^{2k})^2$, then the expected value $\mathbb{E}_{\mathbf{y} \sim \mathcal{N}(0, \mathbf{I})} \frac{\beta_i y_i^2}{\beta_i y_i^2 + \sum_{l \neq i} \beta_l y_l^2} = \lambda_i(k)$ can be expressed as $\mathbb{E}(x_i)$ over random variable $x_i = \frac{u}{u+v_i}$ for $u \sim \mathcal{G}(\frac{1}{2}, 2)$ and $v_i \sim \mathcal{G}(\alpha_i, 2\gamma_i)$ ($\mathcal{G}$ is Gamma distr.) with $\alpha_i = \frac{1}{2} \frac{(\sum_{l \neq i} \beta_l)^2}{\sum_{l \neq i} \beta_l^2}$ and $\gamma_i = \frac{1}{\beta_i} \frac{\sum_{l \neq i} \beta_l^2}{\sum_{l \neq i} \beta_l}$. As PDF $x_i^\bullet(z) = \frac{\gamma_i}{(1-(1-z)\gamma_i)^2} \cdot \mathcal{B}(\frac{\gamma_i z}{1-(1-z)\gamma_i}; \frac{1}{2}, \alpha_i)$ where $\mathcal{B}$ is the Beta distribution and $x_i^\bullet(z)$ enjoys the support $z \in [0; 1]$, then $\lambda_i = \mathbb{E}(z) = \int_0^1 z \cdot x_i^\bullet(z) dz = \gamma_i \frac{\frac{1}{2} \Gamma(\frac{3}{2}) \Gamma(\frac{1}{2} + \alpha_i)}{\Gamma(\frac{1}{2}) \Gamma(\frac{3}{2} + \alpha_i)} \cdot {}_2F_1(\frac{3}{2}, \frac{1}{2} + \alpha_i, \frac{3}{2} + \alpha_i, 1 - \gamma_i)$ where ${}_2F_1(\cdot)$ is the so-called Hypergeometric function.

Proof. See ?? (supplementary material). $\square$

Proposition 3. Analytical Variance. Following assumptions of Proposition 2, the variance ω_i^2 of $x_i^\bullet(z)$ can be expressed as $\omega_i^2 = \mathbb{E}(z^2) - (\mathbb{E}(z))^2 = \int_0^1 z^2 \cdot x_i^\bullet(z) dz - \lambda_i^2 = 0.56419 \gamma_i^{\frac{1}{2}} \frac{\Gamma(\frac{1}{2} + \alpha_i)}{\Gamma(\alpha_i)} (0.4 \cdot {}_2F_1(\frac{5}{2}, 1 - \alpha_i, \frac{7}{2}, 1) \cdot {}_2F_1(\frac{5}{2}, \frac{3}{2} + \alpha_i, \frac{5}{2} + \alpha_i, 1 - \gamma_i) + 0.28571(\gamma_i - 1) \cdot {}_2F_1(\frac{7}{2}, 1 - \alpha_i, \frac{9}{2}, 1) \cdot {}_2F_1(\frac{7}{2}, \frac{3}{2} + \alpha_i, \frac{7}{2} + \alpha_i, 1 - \gamma_i)) - \lambda_i^2$.

Proof. See ?? (supplementary material). $\square$

Note on Spectrum Rebalancing. Fig. 3 explains the consequences of Prop. 2 & 3 and connects them with Alg. 1. Notice following: (i) For $k = 1$ (iterations), the analytical form (Fig. 3b) and the simulated incomplete power iteration (Fig. 3c) both indeed enjoy flatten ϕ and ϕ' for $1 \leq \sigma_1 \leq 3$. (ii) The injected variance is clearly visible in that flattened range (we know the quantity of injected noise). (iii) The analytical and simulated variances match.

Choice of Number of Iterations (k). In Fig. 3b, we plot $\phi(\sigma_i; k)$ using our analytical formulation. The best rebalancing effect is achieved for $k = 1$. For example, the red line ($k = 1$) is mostly flat for σ_i . This indicates the singular values $\sigma_i \geq 1$ are mapped to a similar value which promotes flattening of spectrum. When $\sigma_i \geq 2$ the green line eventually reduces to zero. This indicates that only datasets with spectrum falling into range between 1 and 1.2 will benefit from flattening. Important is to notice also that spectrum augmentation (variance) in Fig. 3b is highest for $k = 1$. Thus, in all experiments (including image classification), we set $k = 1$, and the SFA becomes:

$$\tilde{\mathbf{H}} = \mathbf{H} - \mathbf{H}_{\text{LowRank}} = \mathbf{H} \left(\mathbf{I} - \frac{\mathbf{H}^\top \mathbf{H} \mathbf{r}^{(0)} \mathbf{r}^{(0)\top} \mathbf{H}^\top \mathbf{H}}{\|\mathbf{H}^\top \mathbf{H} \mathbf{r}^{(0)}\|_2^2} \right) \quad (4)$$

3.2 Why does the Incomplete Power Iteration work?

Having discussed SFA, below we show how SFA improves the alignment/generalization by flattening large and boosting small singular values due to rebalanced spectrum.

Improved Alignment. SFA rebalances the weight penalty (by rebalancing singular values) on orthonormal bases, thus improving the alignment of two correlated views. Consider the alignment part of Eq. (2) and ignore the projection head θ for brevity. The contrastive loss (temperature $\tau > 0$, n nodes) on $\mathbf{H}^\alpha$ and $\mathbf{H}^\beta$ maximizes the alignment of two views:

$$\begin{aligned} \mathcal{L}_a &= \mathbb{E}_{\mathbf{h}^\alpha, \mathbf{h}^\beta} (\mathbf{h}^{\alpha\top} \mathbf{h}^\beta / \tau) = \frac{1}{n\tau} \text{Tr}(\mathbf{H}^{\alpha\top} \mathbf{H}^\beta) \\ &= \frac{1}{n\tau} \sum_{i=1}^{d_h} (\sigma_i^\alpha \mathbf{v}_i^{\alpha\top} \mathbf{v}_i^\beta) (\sigma_i^\beta \mathbf{u}_i^{\alpha\top} \mathbf{u}_i^\beta). \end{aligned} \quad (5)$$

The above equation indicates that for $\sigma_i^\alpha \geq 0$ and $\sigma_i^\beta \geq 0$, the maximum is reached if the right and left singular value matrices are perfectly aligned, i.e., $\mathbf{U}^\alpha = \mathbf{U}^\beta$ and $\mathbf{V}^\alpha = \mathbf{V}^\beta$. Notice the singular values σ_i^α and σ_i^β serve as weights for the alignment of singular vectors. As the singular value gap $\Delta\sigma_{12} = \sigma_1 - \sigma_2$ is usually significant (spectrum of feature maps usually adheres to the power law $ai^{-\kappa}$ (i is the index of sorted singular values, a and κ control the magnitude/shape), the large singular values tend to dominate the optimization. Such an issue makes Eq. (5) focus only on aligning the direction of dominant singular vectors, while neglecting remaining

singular vectors, leading to a poor alignment of the orthonormal bases. In contrast, SFA alleviates this issue. According to Prop. 1, Eq. (5) with SFA becomes:

$$\begin{aligned}\mathcal{L}_a^* &= \mathbb{E}_{\mathbf{h}^\alpha, \mathbf{h}^\beta / \tau \mathbf{r}^\alpha, \mathbf{r}^\beta \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} (\tilde{\mathbf{h}}^\alpha \top \tilde{\mathbf{h}}^\beta) \\ &= \frac{1}{n\tau} \sum_{i=1}^{d_h} (1 - \lambda_i^\alpha) \sigma_i^\alpha (\mathbf{v}_i^\alpha \top \mathbf{v}_i^\beta) (1 - \lambda_i^\beta) \sigma_i^\beta (\mathbf{u}_i^\alpha \top \mathbf{u}_i^\beta).\end{aligned}\quad (6)$$

Eq. (6) shows that SFA limits the impact of leading singular vectors on the alignment step if SFA can rebalance the spectra. Indeed, Figure 3b shows the spectrum balancing effect and one can see that $\phi(\sigma_i; k) = \sigma_i(1 - \lambda_i(k)) \leq \sigma_i$. The same may be concluded from $0 \leq \lambda_i \leq 1$ in Prop. 2. See the ?? section (supplementary material) for the estimated upper bound of ϕ . Finally, the [Analysis on Spectral Feature Augmentation](#) section also shows empirically that SFA leads to a superior alignment.

Improved Alignment Yields Better Generalization Bound.

To show SFA achieves the improved generalization bound we quote the following theorem (?).

Theorem 1. *Given a Nearest Neighbour classifier G_f , the downstream error rate of G_f is $\text{Err}(G_f) \leq (1 - \sigma) + R_\varepsilon$, where $R_\varepsilon = P_{\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{A}(\mathbf{x})} \{\|f(\mathbf{x}_1) - f(\mathbf{x}_2)\| \geq \varepsilon\} \leq \frac{\sqrt{2-2\mathcal{L}_a}}{\varepsilon}$, σ is the parameter of the so-called (σ, δ) -augmentation (for each latent class, the proportion of samples located in a ball with diameter δ is larger than σ , $\mathcal{A}(\cdot)$ is the set of augmented samples, $f(\cdot)$ is the encoder, and $\{\|\cdot\| \geq \varepsilon\}$ is the set of samples with ε -close representations among augmented data.*

Proof. See ?? (supp. material). $\square$

Theorem 1 says the key to better generalization of contrastive learning is better alignment $\|f(\mathbf{x}_1) - f(\mathbf{x}_2)\|$ of positive samples. SFA improves alignment by design. See [Why does the Incomplete Power Iteration work?](#) See empirical result in the [Analysis on Spectral Feature Augmentation](#) section. Good alignment (e.g., Fig. 6) due to spectrum rebalancing (e.g., Fig. 5) enjoys $\mathcal{L}_a \leq \mathcal{L}_a^*$ (Eq. (5) and (6)) so one gets $R_\varepsilon^* \leq R_\varepsilon$ and the lower generalization bound $\text{Err}(G_f^*) \leq \text{Err}(G_f)$. Asterisk * means SFA is used ($\mathcal{L}_a^*$ replaces $\mathcal{L}_a$).

4 Experiments

Below we conduct experiments on the node classification, node clustering, graph classification and image classification. For fair comparisons, we use the same experimental setup as the representative Graph SSL (GSSL) methods (i.e., GCA (?) and GRACE (?)).

Datasets. We use five popular datasets (???), including citation networks (Cora, CiteSeer) and social networks (Wiki-CS, Amazon-Computers, Amazon-Photo) (????). For graph classification, we use NC11, PROTEIN and DD (?). For image classification we use CIFAR10/100 (?) and ImageNet-100 (?). See the ?? section for details (supplementary material).

Baselines. We focus on three groups of SSL models. The first group includes traditional GSSL, i.e., Deepwalk (?), node2vec (?), and GAE (?). The second group is

contrastive-based GSSL, i.e., Deep Graph Infomax (DGI) (?), Multi-View Graph Representation Learning (MVGRL) (?), GRACE (?), GCA (?), (?) SUGRL (?). The last group does not require explicit negative samples, i.e., Graph Barlow Twins(G-BT) (?) and BGRL (?). We also compare SFA with COSTA (?) (GCL with feat. augmentation).

Evaluation Protocol. We adopt the evaluation from (???). Each model is trained in an unsupervised manner on the whole graph with node features. Then, we pass the raw features into the trained encoder to obtain embeddings and train an ℓ_2 -regularized logistic regression classifier. Graph Datasets are randomly divided into 10%, 10%, 80% for training, validation, and testing. We report the accuracy with mean/standard deviation over 20 random data splits.

Implementation details We use Xavier initialization for the GNN parameters and train the model with Adam optimizer. For node/graph classification, we use 2 GCN layers. The logistic regression classifier is trained with 5,000 (guaranteed converge). We also use early stopping with a patience of 20 to avoid overfitting. We set the size of the hidden dimension of nodes to from 128 to 512. In clustering, we train a k-means clustering model. For the chosen hyper-parameters see Section ?. We implement the major baselines using PyGCL (?). The detailed settings of augmentation and contrastive objectives are in Table ?? of Section ?.

4.1 Main Results

Node Classification³. We employ node classification as a downstream task to showcase SFA. The default contrastive objective is InfoNCE or the BT loss. Table 1 shows that SFA consistently achieves the best results on all datasets. Notice that the graph-augmented GSSL methods, including GSSL with SFA, significantly outperform the traditional methods, illustrating the importance of data augmentation in GSSL. Moreover, we find that the performance of GCL methods (i.e., GRACE, GCA, DGI, MVGRL) improves by a large margin when integrating with SFA (e.g., major baseline, GRACE, yields 3% and 4% gain on Cora and Citeseer), which shows that SFA is complementary to graph augmentations. SFA also works with the BT loss and improves its performance.

Graph Classification. By adopting the graph-level GNN encoder, SFA can be used for the graph-level pre-training. Thus, we compare SFA with graph augmentation-based models (i.e., GarphCL (?), JOAO (?)) and augmentation-free models (i.e., SimGrace (?), LP-Info (?)). Table 6 shows that SFA outperforms all baselines on the three datasets.

Image Classification⁴. As SFA perturbs the spectrum of feature maps, it is also applicable to the image domain. Table 5 presents the top-1 accuracy on CIFAR10/100 and ImageNet-100. See also the ?? (supp. material).

Runtimes. Table 2 and Fig. 4 show SFA incurs a negligible runtime overhead (few matrix-matrix(vector) multiplications). The cost of SFA for $k < 8$ iterations is negligible.

³Implementation and evaluation are based on PyGCL (?): <https://github.com/PyGCL/PyGCL>.

⁴Implementation/evaluation are based on Solo-learn (?): <https://github.com/vturrissi/solo-learn>.

Method	WikiCS	Am-Comput.	Am-Photo	Cora	CiteSeer	PubMed
RAW fatures	71.98 ± 0.00	73.81 ± 0.00	78.53 ± 0.00	64.61 ± 0.22	65.77 ± 0.15	82.02 ± 0.26
DeepWalk	74.35 ± 0.06	85.68 ± 0.06	89.44 ± 0.11	74.61 ± 0.22	50.77 ± 0.15	80.11 ± 0.25
DGI	75.35 ± 0.14	83.95 ± 0.47	91.61 ± 0.22	82.15 ± 0.63	69.51 ± 0.52	86.01 ± 0.26
MVGRL	77.52 ± 0.08	87.52 ± 0.11	91.74 ± 0.07	83.11 ± 0.12	73.33 ± 0.03	84.27 ± 0.04
GRACE	78.19 ± 0.48	87.25 ± 0.25	92.15 ± 0.25	83.51 ± 0.25	73.63 ± 0.20	85.51 ± 0.37
GCA	78.35 ± 0.05	87.85 ± 0.31	92.49 ± 0.16	82.89 ± 0.21	72.89 ± 0.13	85.12 ± 0.23
SUGRL	77.72 ± 0.28	88.83 ± 0.23	93.28 ± 0.42	83.47 ± 0.55	73.08 ± 0.45	84.91 ± 0.31
MERIT	77.92 ± 0.43	87.53 ± 0.26	93.12 ± 0.42	84.11 ± 0.65	74.34 ± 0.43	84.12 ± 0.23
BGRL	79.11 ± 0.62	87.37 ± 0.40	91.57 ± 0.44	83.77 ± 0.57	73.07 ± 0.06	84.62 ± 0.35
G-BT	76.85 ± 0.62	86.86 ± 0.33	92.63 ± 0.57	83.63 ± 0.44	72.95 ± 0.17	84.52 ± 0.12
COSTA	79.12 ± 0.02	88.32 ± 0.03	92.56 ± 0.45	84.32 ± 0.22	72.92 ± 0.31	86.01 ± 0.22
SFA _{BT}	80.22 ± 0.05	88.14 ± 0.15	92.83 ± 0.14	84.10 ± 0.01	73.73 ± 0.03	85.63 ± 0.07
SFA _{InfoNCE}	79.98 ± 0.05	89.24 ± 0.27	93.53 ± 0.16	85.89 ± 0.01	75.31 ± 0.03	86.29 ± 0.13

Table 1: Node classification on graph datasets. Note that SFA_{InfoNCE} and SFA_{BT} can be directly compared to GRACE and G-BT. (Accuracy is reported.)

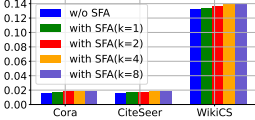


Figure 4: Running time per epoch in seconds.

Datasets	w/o SFA	SFA $k = 1$
Cora	165	172
CiteSeer	164	173
WikiCS	1334	1343

Table 2: Running time per epoch in seconds.

Method	SFA (ours)	SVD	Random SVD
Time	0.25 hour	12 hours	1.2 hours

Table 3: Running time on Ogb-arxiv.

Ogb-arxiv	Validation	Test
MLP	57.6 ± 0.2	55.5 ± 0.3
Node2vec	71.2 ± 0.3	70.0 ± 0.3
MVGLR	69.3 ± 0.3	68.2 ± 0.2
DGI	71.2 ± 0.1	70.3 ± 0.1
SUGRL	70.2 ± 0.1	69.3 ± 0.2
MERIT	67.2 ± 0.1	65.3 ± 0.2
GRACE	71.4 ± 0.5	70.8 ± 0.1
G-BT	71.1 ± 0.3	70.0 ± 0.2
COSTA	71.6 ± 0.4	71.0 ± 0.4
SFA _{Info}	72.3 ± 0.1	71.6 ± 0.4

Table 4: Node classification.

Method	CIFAR10	CIFAR100	ImageNet-100
SimCLR	90.5	65.5	76.8
SFA _{SimCLR}	91.6	66.7	77.7
BalowTw	92.0	69.7	80.0
SFA _{BT}	92.5	70.4	80.9
Siamese	90.51	66.04	74.5
SFA _{Siamese}	91.23	66.99	75.6
SwAV	89.17	64.88	74.0
SFA _{SwAV}	90.12	65.82	74.8

Table 5: Image classification on CIFAR10, CIFAR100 and ImageNet-100.

Method	NCII	PROTEIN	DD
GraphCL	77.8 ± 0.4	74.3 ± 0.4	77.8 ± 0.4
LP-Info	75.8 ± 1.2	74.6 ± 0.2	72.5 ± 1.9
JOAO	78.0 ± 0.4	74.5 ± 0.4	77.5 ± 0.5
SimGRACE	77.4 ± 1.0	73.9 ± 0.1	77.3 ± 1.1
SFA _{InfoNCE}	78.7 ± 0.4	75.4 ± 0.4	78.6 ± 0.4

Table 6: Graph classification. (SFA_{InfoNCE} can be directly compared with GraphCL.)

$\mathcal{A}_G$	$\mathcal{A}_F$	$\mathcal{A}_{F^*}$	Am-Comput.	Cora	CiteSeer
×	×	×	85.01	80.04	71.35
✓	×	×	87.25	82.23	74.56
×	×	✓	86.74	84.19	73.15
✓	×	✓	88.74	85.90	75.05
✓	✓	×	87.55	83.35	74.45
✓	✓	✓	88.69	84.50	74.70

Table 7: Ablation study on different augmentation strategies: $\mathcal{A}_G$, $\mathcal{A}_F$ and $\mathcal{A}_{F^*}$.

To rebalance the spectrum, choices for a push-forward function whose curve gets flatter as singular values get larger are many but they require SVD to rebalance singular values directly. The complexity of SVD is $\mathcal{O}(nd_h^2)$ for $n \gg d_h$, which is costly in GCL as SVD has to be computed per mini-batch (and SVD is unstable in back-prop. due to non-simple singular values). Table 3 shows the runtime on Ogb-arxiv.

4.2 Analysis on Spectral Feature Augmentation

Improved Alignment. We show that SFA improves the alignment of two views during training. Let $\mathbf{H}^\alpha$ and $\mathbf{H}^\beta$ ($\tilde{\mathbf{H}}^\alpha$ and $\tilde{\mathbf{H}}^\beta$) denote the feature maps of two views without (with) applying SFA. The alignment is computed by $\|\mathbf{H}^\alpha - \mathbf{H}^\beta\|_F^2$ (or $\|\tilde{\mathbf{H}}^\alpha - \tilde{\mathbf{H}}^\beta\|_F^2$). Fig. 6 shows that SFA achieves better alignment. See also the ??.

Empirical Evolution of Spectrum. Fig. 5 (Cora) shows how the singular values of features maps $\mathbf{H}$ (without SFA) and $\tilde{\mathbf{H}}$ (with SFA) evolve. As training progresses, the gap between consecutive singular values gradually decreases due to SFA. The leading components of spectrum (where the signal is) become more balanced: empirical results match Prop. 1 & 2.

Ablations on Augmentations. Below, we compare graph augmentation $\mathcal{A}_G$, channel feature augmentation $\mathcal{A}_F$ (random noise added to embeddings directly) and the spectral feature augmentation $\mathcal{A}_{F^*}$. Table 7 shows that using $\mathcal{A}_G$ or $\mathcal{A}_{F^*}$ alone improves performance. For example, $\mathcal{A}_G$ yields 2.2%, 2.2% and 3.2% gain on Am-Computer, Cora, and CiteSeer. $\mathcal{A}_{F^*}$ yields 1.7%, 4.1% and 1.8% gain on Am-Computer, Cora, and CiteSeer. Importantly, when both $\mathcal{A}_G$ and $\mathcal{A}_{F^*}$ are applied, the gain is **3.7%**, **6.0%** and **4.8%** on Am-Computer, Cora, and CiteSeer over “no augmentations”. Thus, SFA is complementary to existing graph augmentations. We also notice that $\mathcal{A}_{F^*}$ with $\mathcal{A}_G$ outperforms $\mathcal{A}_F$ with $\mathcal{A}_G$ by 1.1%, 2.4% and 1.7%, which highlights the benefit of SFA.

Effect of Number of Iterations (k). Below, we analyze how k in Eq. (3) influences the performance. We set $k \in \{0, 1, 2, 4, 8\}$ in our model with the InfoNCE loss on Cora, CiteSeer and Am-Computer. The case of $k = 0$ means that no power iteration is used, *i.e.*, the solution simplifies to the feature augmentation by subtracting from $\mathbf{H}$ perturbation $\mathbf{H}\mathbf{r}^{(0)}\mathbf{r}^{(0)\top}/\|\mathbf{r}^{(0)}\|_2^2$ with random $\mathbf{r}^{(0)}$. Table 8 shows that without power iteration, the performance of the model drops, *i.e.*, 1.5%, 3.3% and 2.2% on Am-Comp., Cora and CiteSeer.

The best gain in Table 8 is achieved for $1 \leq k \leq 2$. This is consistent with Prop. 2, Fig. 3b and 3c, which show that the spectrum balancing effect (approximately flat region of ϕ) and significant spectrum augmentation (indicated by the large deviation in Fig. 3b) are achieved only when the power iteration is incomplete (low k , *i.e.*, $1 \leq k \leq 2$).

Robustness to Noisy Features. Below we check on Cora and CiteSeer if GCL with SFA is robust to noisy node features in node classification setting. We draw noise $\Delta\mathbf{x} \sim \mathcal{N}(0, \mathbf{I})$ and inject it into the original node features as $\mathbf{x} + \epsilon\Delta\mathbf{x}$, where $\epsilon \geq 0$ controls the noise intensity. Table 9 shows that SFA is robust to the noisy features. SFA partially balances the spectrum (Fig 3b and 2b) of leading singular components where the signal lies, while non-leading components where the noise resides are left mostly unchanged by SFA.

Other Variants of Spectral Augmentation. Below, we experiment with other ways of performing spectral augmentation. Figure 7 shows three different push-forward functions: our SFA, MaxExp(F) (?) and Grassman feature maps (?). As MaxExp(F) and Grassman do not inject any spectral noise, we equip them with explicit noise injectors. To determine what is the best form of such an operator, we vary (i) where the noise is injected, (ii) profile of balancing curve. As the push-forward profiles of SFA and MaxExp(F) are similar (Figure 7), we implicitly inject the noise into MaxExp(F)

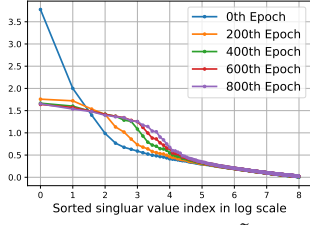


Figure 5: Spectrum of $\tilde{\mathbf{H}}$ when training on Cora with SFA.

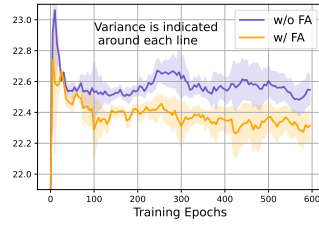


Figure 6: Alignment of $\mathbf{f}$. maps of two views (lower is better).

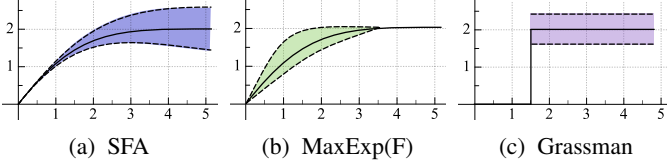


Figure 7: Spectrum augmentation variants (dashed lines: injected noise var.) Power Norm. & Power Norm.* resemble (b) and (a).

along the non-leading singular values (*c.f.* leading singular values of SFA).

$\text{MaxExp}(\mathbf{F})$ is only defined for symmetric-positive definite matrices. We extend it to feature maps by:

$$\tilde{\mathbf{H}} = \mathbf{H}\mathbf{B}(\mathbf{I} - (\mathbf{I} - \mathbf{A}/\text{Tr}(\mathbf{A}))^{\eta+\Delta\eta}), \quad (7)$$

where $\mathbf{A} = (\mathbf{H}^\top \mathbf{H})^{0.5}$ and $\mathbf{B} = (\mathbf{H}^\top \mathbf{H})^{-0.5}$ are obtained via highly-efficient Newton-Schulz iteration. We draw the random noise $\Delta\eta \sim \alpha\mathcal{N}(0, 1)$ for each $\mathbf{H}$ and control the variance by $\alpha > 0$. We round $\eta + \Delta\eta$ to the nearest integer ($\text{MaxExp}(\mathbf{F})$ requires an integer for fast matrix exponentiation) and ensure the integer is greater than zero. See the ?? section for details.

Matrix Square Root is similar to $\text{MaxExp}(\mathbf{F})$ but the soft function flattening the spectrum is replaced with a matrix square root push-forward function. *Power Norm.* yields:

$$\tilde{\mathbf{H}} = \mathbf{H}\mathbf{B} \text{PowerNorm}(\mathbf{A}; \beta + \Delta\beta), \quad (8)$$

where $0 \leq \beta + \Delta\beta \leq 1$ controls the steepness of the balancing push-forward curve. $\Delta\beta$ is drawn from the Normal distribution (we choose the best aug. variance). *Power Norm.** is:

$$\begin{aligned} \tilde{\mathbf{H}} &= \mathbf{H}\mathbf{B}((1 - \beta - \Delta\beta)\mathbf{A}^{0.5} + (\beta + \Delta\beta)\mathbf{A}^{0.5}\mathbf{A}^{0.25}) \\ &= \mathbf{H}\mathbf{B} \text{PowerNorm}(\mathbf{A}; \beta + \Delta\beta), \end{aligned} \quad (9)$$

where $0 \leq 1 + \Delta\beta^* \leq 2$, $\Delta\beta^*$ is drawn from the Normal dist. $\mathbf{A}^{0.5}$ and $\mathbf{A}^{0.25} = (\mathbf{A}^{0.5})^{0.5}$ are obtained by Newton-Schulz iterations. See ?? for details.

Grassman feature maps can be obtained as:

$$\tilde{\mathbf{H}} = \mathbf{H}\mathbf{B}(\mathbf{V}\text{Flat}(\mathbf{A}; \kappa)\mathbf{V}^\top), \quad (10)$$

where $\text{Flat}(\cdot; \kappa)$ sets $\kappa > 0$ leading singular values to $1 + \Delta\kappa$ (the rest is 0), and $\Delta\kappa \sim \alpha\mathcal{N}(0, \mathbf{I})$. In experiments we use SVD and random SVD (injects itself some noise) to produce $\mathbf{V}$ and $\mathbf{A}$ from $\mathbf{A}$. See ?? for details.

Table 10 shows that SFA outperforms $\text{MaxExp}(\mathbf{F})$, Power Norm., Power Norm.* and Grassman. As SFA augments parts of spectrum where signal resides (leading singular values) which is better than augmenting non-leading singular values

Power Iteration	Am-Computer	Cora	CiteSeer
$k = 0$	87.25	83.04	71.35
$k = 1$	88.83	85.90	73.56
$k = 2$	88.72	85.59	75.3
$k = 4$	88.64	85.29	74.8
$k = 8$	88.27	85.23	74.9

Table 8: Results on different k of SFA.

ϵ		10^{-4}	10^{-3}	10^{-2}	10^{-1}
Cora	w/o SFA	80.55	70.67	62.85	59.23
	with SFA	83.14	76.56	69.59	62.55
CiteSeer	w/o SFA	67.48	62.32	52.69	42.81
	with SFA	72.12	70.36	60.40	46.44

Table 9: Results on noisy features.

Spectrum Aug.	Am-Comput.	Cora	CiteSeer
SFA (ours)	88.83	85.90	75.3
MaxExp(F)	87.13	84.19	72.85
MaxExp(F) (w/o noise)	86.83	83.52	72.35
Power Norm.*	86.7	82.9	70.4
Power Norm.	86.6	82.7	70.2
Power Norm. (w/o noise)	86.3	82.5	69.9
Grassman	86.23	82.57	70.15
Grassman (w/o noise)	85.83	81.17	69.85
Grassman (rand. SVD)	85.33	81.01	69.95
Matrix Precond.	82.52	78.14	67.73

Table 10: Results of various spectral augmentations.

(some noise may reside there) as in $\text{MaxExp}(\mathbf{F})$. As Grassman binarizes spectrum, it may reject some useful signal at the boundary between leading and non-leading singular values. Finally, ?? model reduces the spectral gap, thus rebalances the spectrum (also non-leading part).

5 Conclusions

We have shown that GCL is not restricted to only link perturbations or feature augmentation. By introducing a simple and efficient spectral feature augmentation layer we achieve significant performance gains. Our incomplete power iteration is very fast. Our theoretical analysis has demonstrated that SFA rebalances the useful part of spectrum, and also augments the useful part of spectrum by implicitly injecting the noise into singular values of both data views. SFA leads to a better alignment with a lower generalization bound.

6 Acknowledgments

We thank anonymous reviewers for their valuable comments. The work described here was partially supported by grants from the National Key Research and Development Program of China (No. 2018AAA0100204) and from the Research Grants Council of the Hong Kong Special Administrative Region, China (CUHK 2410021, Research Impact Fund, No. R5034-18). Minus temporibus veritatis adipisci quia blanditiis a nisi earum tenetur, necessitatibus molestias iure quas incident aliquam sunt in, possimus placeat nostrum voluptatum, repellendus commodi omnis debitis qui pariat officis rem, beatae nam cupiditate sunt ut. Itaque unde vitae a, quam quibusdam repellat quae laboriosam neque ullam voluptatem pariat tenetur, aperiam molestiae veritatis laudantium quos deleniti explicabo, eveniet consequuntur quidem provident voluptas neque quibusdam totam distinctio iste deserunt, ipsa blanditiis aliquid corporis? Beatae commodi eum dolores alias

temporibus, magni expedita id nam sint autem corporis quia voluptates ipsam suscipit, natus in sapiente veritatis officia vero vel atque fuga, dolor esse quae mollitia reprehenderit.